

SQL PROJECT REPORT

Employee Management System

Database Design • Data Cleaning • SQL Analytics

PROJECT SCOPE

A MySQL-based relational database project for managing employee, job/department, salary & bonus, qualification, leave, and payroll information, followed by business-oriented SQL analysis.

KPI	Value
Employees	60
Departments	8
Database Tables	6
Qualification Records	60
Leave Records	60
Payroll Records	60

Prepared as a portfolio / academic project report

1. Executive Summary

The Employee Management System is a relational SQL project built in MySQL to demonstrate an end-to-end workflow: database creation, relational schema design, CSV data loading, data inspection and cleaning, and business-oriented SQL analysis. The project contains six interconnected tables covering departments and roles, salary structures, employees, qualifications, leaves, and payroll.

The supplied dataset contains 60 employees across 8 departments. The average employee age is 35.15 years, with 31 male and 29 female records. Payroll contains 60 April 2024 records, totaling ₹2,778,000, with an average of ₹46,300 per record.

KEY OUTCOME

The project demonstrates practical SQL skills beyond basic SELECT statements, including multi-table JOINS, aggregation, CTEs, subqueries, window functions, string/date operations, and data-quality checks.

1.1 Key Findings at a Glance

- Finance and IT have the largest employee/role presence, with 9 records each.
- Legal has the highest average base salary at approximately ₹84,600, followed by Engineering at ₹81,143.
- Finance Director is the highest-paid role at ₹170,000 base salary with a ₹35,000 bonus.
- All 60 employees have a qualification record.
- All 60 employees have one leave record in the supplied dataset.
- April 2024 payroll totals ₹2,778,000; average payroll amount is ₹46,300.

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3. Project Objectives and Scope

3.1 Objectives

- Design a relational database for employee-management data.
- Create tables with primary keys and foreign-key relationships.
- Load CSV files into MySQL using LOAD DATA INFILE.
- Inspect and clean text, numeric, date, and identifier fields using SQL.
- Use joins and aggregations to answer workforce, salary, qualification, leave, and payroll questions.
- Demonstrate CTEs, subqueries, RANK(), DENSE_RANK(), PARTITION BY, GROUP BY, HAVING, and date/string functions.
- Translate query results into business-oriented HR and compensation insights.

3.2 Scope

The project is an analytical employee-management database rather than a complete production HR application. Its scope is centered on relational data modeling and SQL analysis; it does not include a user interface, production authentication, automated ETL orchestration, or live payroll integration.

4. Technology and Project Structure

Component	Technology / Asset	Purpose
Database	MySQL 8.x	Relational storage, schema and queries
Query Language	SQL	DDL, loading, cleaning and analysis
Input Data	CSV	Six source datasets
Data Model	ER Diagram	Visual entity/relationship model
Documentation	SQL scripts + presentation	Reproducible project workflow

5. Database Architecture and ER Model

The database is named employee_management and contains six core tables. Primary keys uniquely identify records, while foreign keys connect employee, job, salary, qualification, leave, and payroll information.

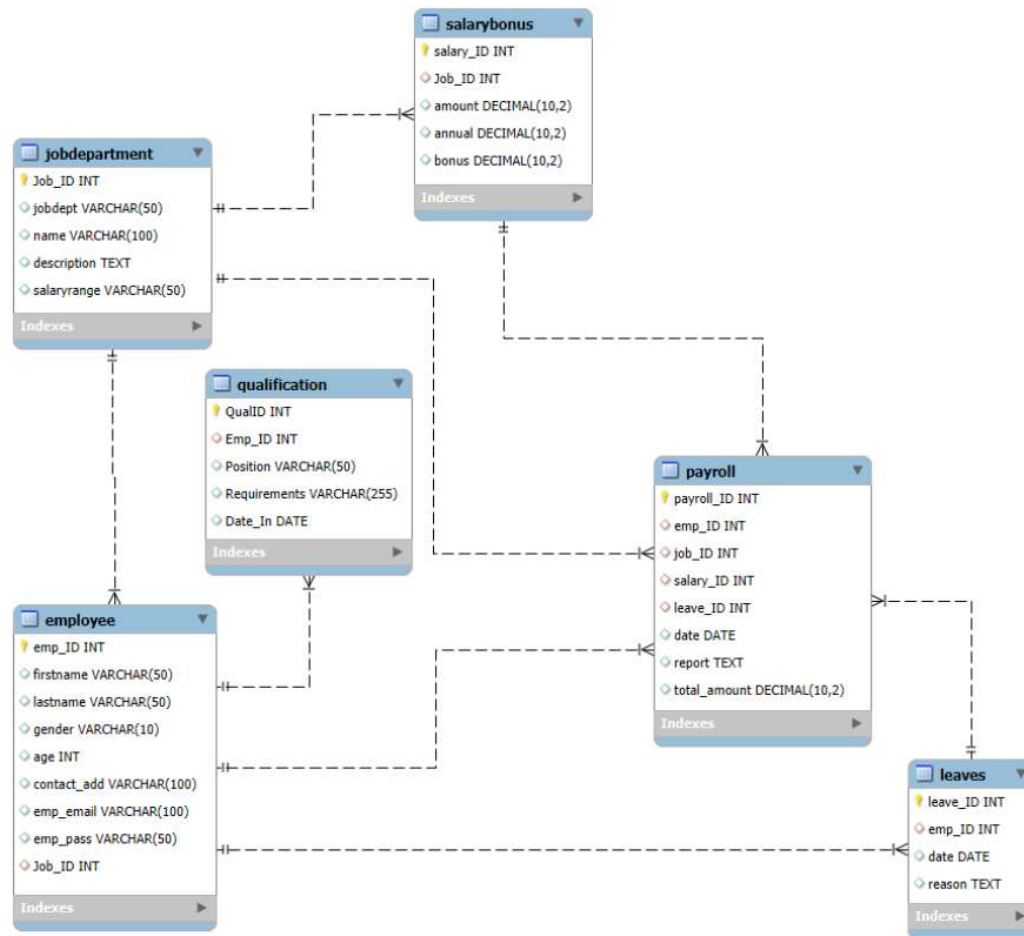


Figure 1. Entity Relationship Diagram supplied with the project.

5.1 Table Dictionary

Table	Primary Key	Business Purpose
JobDepartment	Job_ID	Department/role master with job name, description and salary range.
SalaryBonus	salary_ID	Salary, annual salary and bonus associated with Job ID.
Employee	emp_ID	Employee identity/contact details and assigned Job ID.
Qualification	QualID	Qualification/requirement information linked to employees.
Leaves	leave_ID	Employee leave events with date and reason.
Payroll	payroll_ID	Payroll transaction/report linked to employee, job, salary and leave.

5.2 Relationship Design

- JobDepartment → Employee: Employee.Job_ID references JobDepartment.Job_ID; deletion sets Job_ID to NULL.
- JobDepartment → SalaryBonus: SalaryBonus.Job_ID references JobDepartment.Job_ID.

- Employee → Qualification: Qualification.Emp_ID references Employee.emp_ID with cascading deletion.
- Employee → Leaves: Leaves.emp_ID references Employee.emp_ID with cascading deletion.
- Payroll links employee, job, salary and leave records through foreign keys.
- Referential actions such as CASCADE and SET NULL are used to maintain relationship integrity.

6. Data Sources and Dataset Profile

The project uses six CSV files. Each file contains 60 records, producing a compact relational dataset suitable for demonstrating SQL joins, aggregations, cleaning and analytical patterns.

Dataset	Rows	Columns	Main Fields
Employee	60	9	EmplID, name, gender, age, email, JobID
JobDepartment	60	5	JobID, department, role name, salary range
Salary_Bonus	60	5	SalaryID, JobID, amount, annual, bonus
Qualification	60	5	QualID, EmplID, position, requirements, Date In
Leaves	60	4	LeaveID, EmplID, date, reason
Payroll	60	8	PayrollID, EmplID, JobID, SalaryID, LeaveID, date, total amount

6.1 Workforce Profile

Metric	Result
Total employees	60
Departments	8
Average age	35.15 years
Age range	25–55 years
Male records	31 (51.7%)
Female records	29 (48.3%)
Qualification records	60
Leave records	60
Payroll records	60

7. Data Loading Workflow

data_loading.sql checks MySQL's secure_file_priv setting and then imports each CSV with LOAD DATA INFILE, using comma delimiters, quoted fields, and IGNORE 1 ROWS for the header.

```
SHOW VARIABLES LIKE 'secure_file_priv';
```

```
LOAD DATA INFILE '.../Employee.csv'
INTO TABLE Employee
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

The original script uses a local Windows MySQL upload directory. When executed elsewhere, the file path must be adapted to the permitted `secure_file_priv` directory or replaced with another supported loading method.

8. Data Cleaning and Quality Checks

The cleaning workflow is table-by-table: schema inspection, NULL checks, text normalization, email standardization, duplicate checks and numeric validation.

Table	Cleaning / Validation	Purpose
JobDepartment	NULL checks; remove '\$' from salary range; TRIM text	Standardize and enable salary-range parsing
SalaryBonus	NULL checks; negative salary/bonus validation	Validate numeric compensation fields
Employee	NULL checks; TRIM; lowercase/remove email spaces; duplicate check	Standardize contact data
Qualification	Data type and NULL checks	Inspect required fields
Leaves	Data type and NULL checks; TRIM reason	Standardize leave reason
Payroll	Data type and NULL checks; TRIM report	Standardize payroll text

8.1 Data Quality Observations

- All six CSV datasets contain 60 records.
- Employee emails are unique in the supplied data and are normalized to lowercase by the cleaning script.
- `JobDepartment.salaryrange` is stored as text and contains '\$' symbols; the script removes the symbols before parsing.
- Salary and bonus values are checked for negative values.
- The employee password field contains repeated placeholder-style values. This is acceptable for a demo dataset but not for production; passwords should never be stored in plaintext.
- Qualification and leave records are one per employee, so those dimensions are uniform rather than representative of complex real-world HR behavior.

9. SQL Analysis and Business Questions

The analysis script is organized into five business sections: employee insights; job role and department analysis; qualification and skills analysis; leave and absence patterns; and payroll/compensation analysis.

9.1 Employee Insights

- Workforce size: 60 employees.
- Finance and IT are the largest departments, with 9 and 9 employees respectively.
- Salary-table expenditure: ₹4,321,000.
- Finance Director is the highest-paid role.

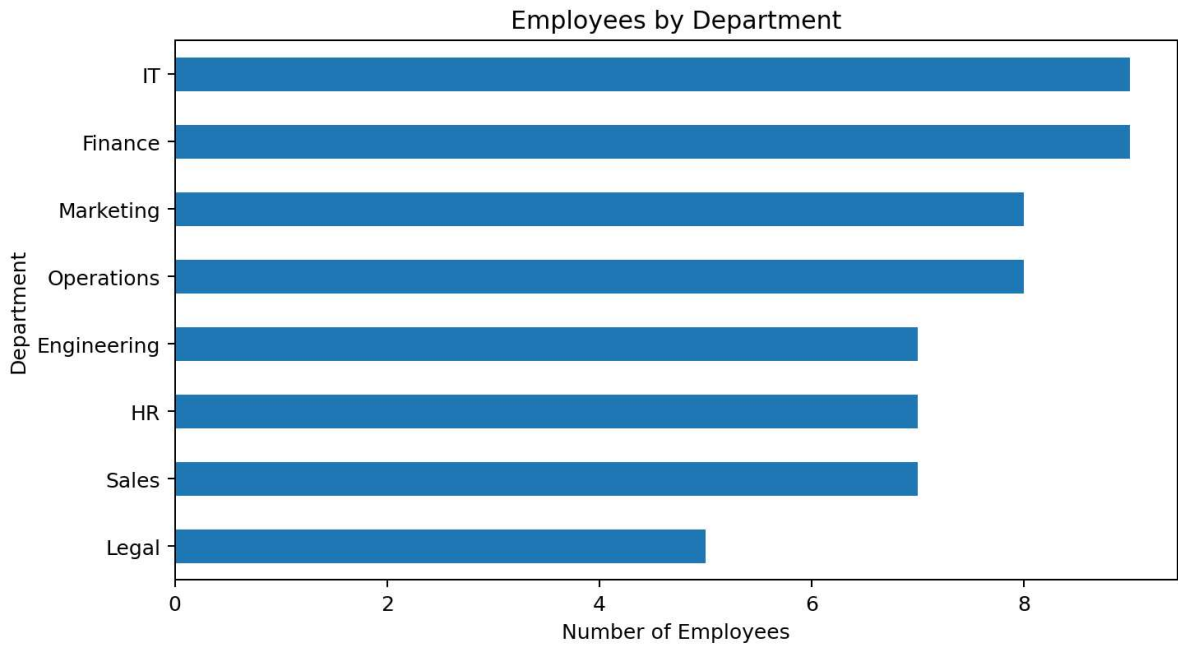


Figure 2. Employee distribution across departments.

9.2 Job Role and Department Analysis

Department	Roles	Avg Salary	Salary Allocation	Total Bonus
Finance	9	₹72,333	₹651,000	₹96,000
IT	9	₹70,889	₹638,000	₹94,000
Operations	8	₹68,750	₹550,000	₹77,500
Marketing	8	₹65,625	₹525,000	₹73,000
Engineering	7	₹81,143	₹568,000	₹88,000
Sales	7	₹75,429	₹528,000	₹78,500
HR	7	₹62,571	₹438,000	₹57,200
Legal	5	₹84,600	₹423,000	₹66,500

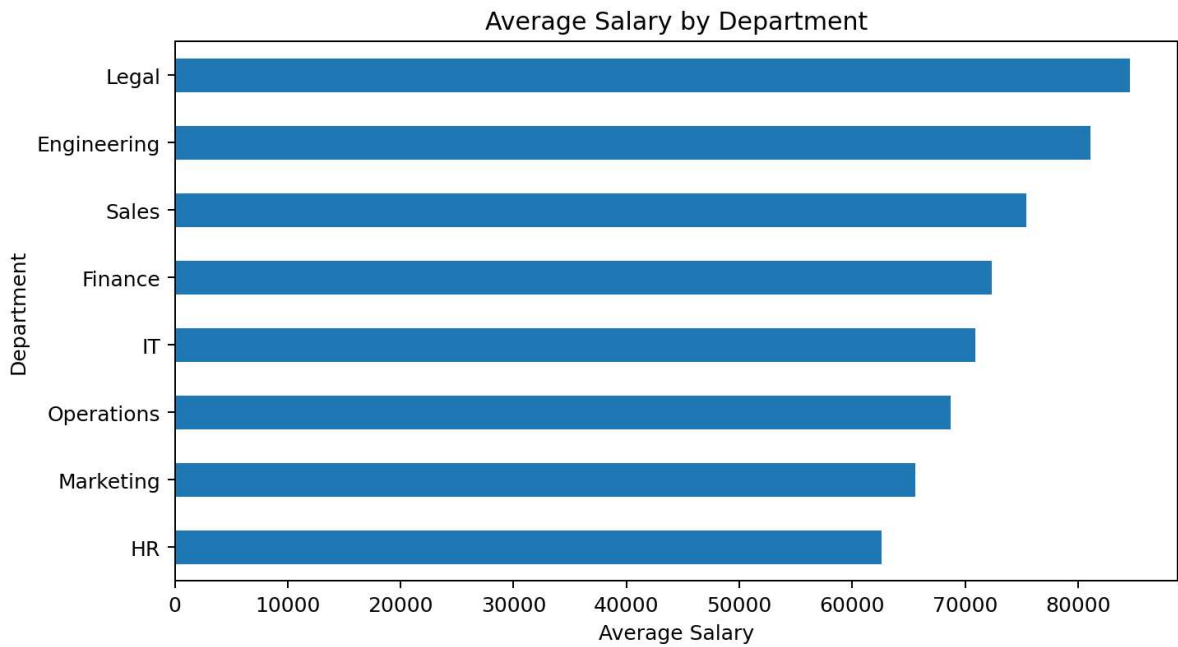


Figure 3. Average salary by department.

9.3 Qualification and Skills Analysis

- 60 unique employees have qualification records.
- There is one qualification record per employee in the supplied dataset.
- Because qualification counts do not vary, the current sample cannot meaningfully identify employees with substantially more qualifications.

9.4 Leave and Absence Patterns

Metric	Result
Total leave records	60
Leave year represented	2024
Average leaves per employee	1.00
Employees with leave records	60
Most common leave reason	Travel
Frequency of most common reason	3

The analysis also compares leave counts with payroll amounts. Since every employee has one leave record in this dataset, the sample is not suitable for establishing a strong relationship between leave frequency and pay.

9.5 Payroll and Compensation Analysis

Payroll Metric	Value
Payroll period	April 2024
Total payroll processed	₹2,778,000
Average payroll amount	₹46,300
Minimum payroll amount	₹38,500
Maximum payroll amount	₹51,500
Total base salary in SalaryBonus	₹4,321,000

Average bonus	₹10,512
Maximum bonus	₹35,000

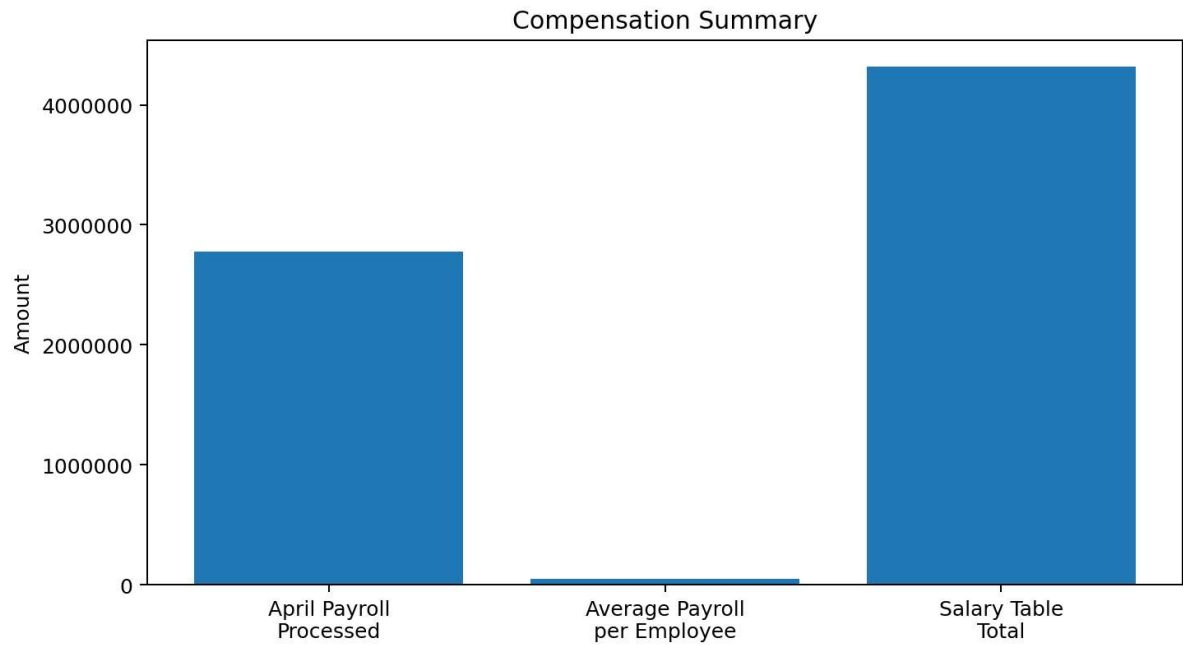


Figure 4. Compensation values in the supplied data.

10. Advanced SQL Techniques Demonstrated

Technique	Application in Project
JOINS	Combines employee, department, salary, leave and payroll data.
GROUP BY + aggregates	COUNT(), SUM(), AVG(), MAX() for HR and compensation summaries.
HAVING	Filters grouped results, including maximum-leave analysis.
String functions	TRIM(), LOWER(), REPLACE(), SUBSTRING_INDEX() for cleaning/parsing.
Date functions	YEAR() and DATE_FORMAT() for leave and payroll periods.
CTEs	WITH clauses for top-paid and above-average salary analysis.
Window functions	RANK(), DENSE_RANK(), and PARTITION BY for salary ranking.
Subqueries	Nested AVG() and MAX() logic for above-average and maximum-leave analysis.

10.1 Representative Query: Department Salary Ranking

```
SELECT jd.jobdept,
       e.emp_ID,
       sb.amount,
       RANK() OVER (
         PARTITION BY jd.jobdept
         ORDER BY sb.amount DESC
       ) AS dept_rank
FROM Employee e
JOIN JobDepartment jd
  ON e.Job_ID = jd.Job_ID
```

```
JOIN SalaryBonus sb
ON jd.Job_ID = sb.Job_ID;
```

This query ranks employees within their own department instead of producing only a company-wide ranking.

10.2 Representative Query: Above-Average Salary

```
SELECT e.emp_ID,
       e.firstname,
       sb.amount
FROM Employee e
JOIN SalaryBonus sb
ON e.Job_ID = sb.Job_ID
WHERE sb.amount > (
    SELECT AVG(amount)
    FROM SalaryBonus
);
```

11. Key Analytical Insights

Area	Insight
Workforce Structure	Finance and IT each have 9 records, the highest department counts.
Compensation	Legal has the highest department average salary at approximately ₹84,600.
Top Role	Finance Director has the highest base salary at ₹170,000.
Bonus Allocation	Finance has the highest total bonus allocation at ₹96,000.
Payroll	April 2024 payroll totals ₹2,778,000, averaging ₹46,300.
Qualifications	Every employee has a qualification record in the supplied data.
Leave Pattern	Every employee has one leave record, so absence frequency is uniform in this sample.

12. Data Model Considerations and Limitations

- SalaryBonus is keyed by Job_ID, so salary is modeled at the job level rather than as employee-specific compensation history. A production design would typically add an employee compensation/history table.
- JobDepartment combines department and job-role information. A more normalized design could separate Department and JobRole.
- Payroll links to Leave through leave_ID. Production payroll would normally represent deductions and attendance adjustments with explicit transaction and payroll-period rules.
- Payroll contains a single period (April 2024), limiting trend and seasonality analysis.
- Exactly one leave record per employee prevents meaningful absenteeism distribution analysis.
- salaryrange is stored as text; separate numeric minimum_salary and maximum_salary columns would improve analytical reliability.
- emp_pass is present in the Employee table. Real systems should use salted password hashes or a dedicated identity provider and should not expose credentials in analytical tables.
- The dataset appears synthetic/demo-oriented; findings should be treated as SQL-analysis demonstrations rather than real HR decisions.

13. Recommendations and Future Enhancements

- Separate Department and JobRole entities for stronger normalization.
- Create EmployeeSalaryHistory with effective dates for promotions and increments.
- Add Attendance and LeaveBalance tables for requests, approvals, balances and payroll deductions.
- Create a PayrollPeriod structure and separate gross pay, deductions, taxes, bonuses and net pay.
- Remove plaintext password storage and implement secure authentication outside the analytical schema.
- Add CHECK constraints for valid age, non-negative compensation and controlled categorical values.
- Add indexes on frequently joined foreign keys such as Job_ID and Emp_ID.
- Expand the data across multiple months/years for trend analysis.
- Create views/stored procedures for recurring HR reports.
- Connect the database to Power BI for an interactive HR dashboard.

14. Conclusion

The Employee Management System successfully demonstrates a practical SQL workflow from relational database design to analytical reporting. It covers six related entities, CSV-based data ingestion, systematic cleaning, referential integrity, and a broad set of analytical queries.

Its strongest portfolio value is the breadth of SQL techniques demonstrated: multi-table joins, aggregations, string/date functions, CTEs, subqueries and window functions. The project also connects technical query results with business questions, which is an important skill for data and BI roles.

PORTFOLIO VALUE

A strong next step would be to add multi-period HR transactions, strengthen historical compensation modeling, and build a Power BI dashboard on top of the cleaned analytical layer.

15. Appendix: SQL Inventory and Representative Queries

15.1 Included SQL Scripts

Script	Purpose
database_&_table_creation.sql	Creates employee_management and six tables with primary/foreign keys.
data_loading.sql	Loads the six CSV datasets with LOAD DATA INFILE.
data_cleaning.sql	Performs inspection, NULL checks, standardization and validation.
data_analysis.sql	Contains business questions, insights, CTEs, window functions, subqueries and bonus analysis.

15.2 Representative Database Definition

```
CREATE TABLE Employee (
    emp_ID INT PRIMARY KEY,
    firstname VARCHAR(50),
    lastname VARCHAR(50),
    gender VARCHAR(10),
    age INT,
    contact_add VARCHAR(100),
    emp_email VARCHAR(100) UNIQUE,
    emp_pass VARCHAR(50),
    Job_ID INT,
    CONSTRAINT fk_employee_job
        FOREIGN KEY (Job_ID)
        REFERENCES JobDepartment(Job_ID)
        ON DELETE SET NULL
        ON UPDATE CASCADE
);
```

15.3 Representative Cleaning Operations

```
UPDATE Employee
SET firstname = TRIM(firstname),
    lastname = TRIM(lastname),
    gender = TRIM(gender),
    emp_email = TRIM(emp_email);

UPDATE Employee
SET emp_email = LOWER(emp_email);

SELECT emp_email, COUNT(*)
FROM Employee
GROUP BY emp_email
HAVING COUNT(*) > 1;
```

15.4 Representative CTE

```
WITH salary_data AS (
    SELECT e.emp_ID, e.firstname, e.lastname, sb.amount
    FROM Employee e
    JOIN SalaryBonus sb ON e.Job_ID = sb.Job_ID
)
SELECT *
FROM salary_data
ORDER BY amount DESC
LIMIT 5;
```

15.5 Project Metrics Used in This Report

Metric	Value
Employees	60
Departments	8
JobDepartment records	60
SalaryBonus records	60
Qualification records	60
Leave records	60
Payroll records	60
Salary table total	₹4,321,000
April 2024 payroll total	₹2,778,000
Average payroll amount	₹46,300
Highest base salary	₹170,000
Highest bonus	₹35,000

— End of Report —